
REPASO DE CONCEPTOS BÁSICOS DE ESTADÍSTICA

Carlos Fernando Vásquez Guerra

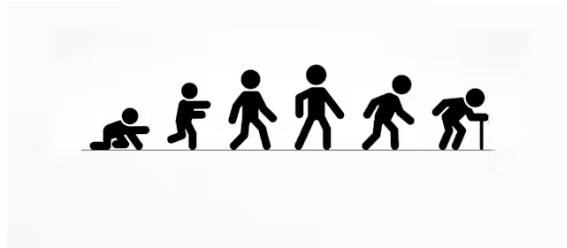


DATOS CUANTITATIVOS

DATOS CUALITATIVOS

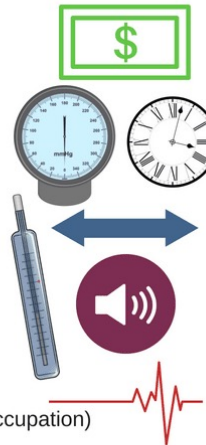
Grace LaConte's

Quantitative and Qualitative Data Examples



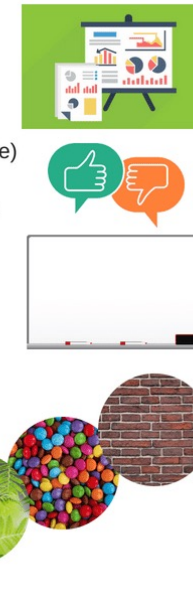
Quantitative Data

- money
- time
- speed
- movement
- height
- length
- area
- volume
- weight
- temperature
- humidity
- pressure
- sound level
- categories (age, gender, occupation)
- positioning
- status



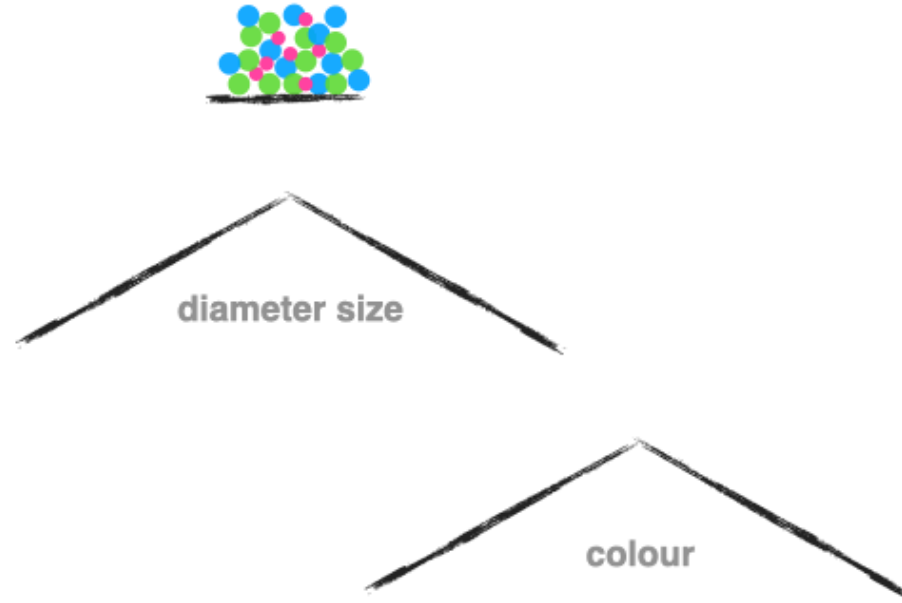
Qualitative Data

- verbal and written feedback
- narrative story
 - first-hand (direct experience)
 - second-hand (telling someone else)
 - third-hand (outside story-teller)
- visual images, drawings, or models
- experiential sensations
- descriptions of
 - colors
 - textures
 - smells
 - tastes
 - appearance
 - beauty
 - feelings
 - intuition
 - sensations
 - choices
 - values
 - beliefs



APLICACIÓN EN DATA ANALYTICS

COMBINACIÓN DE DATOS



APLICACIÓN EN DATA ANALYTICS

TRANSFORMACIÓN DE DATOS CUALITATIVOS A CUANTITATIVOS

ONE-HOT ENCODING

Original Data						
Team	Points					
Team	Points	Team_A	Team_B	Team_C	Points	
A	25	1	0	0	25	
A	12	1	0	0	12	
B	15	0	1	0	15	
B	14	0	1	0	14	
B	19	0	1	0	19	
B	23	0	1	0	23	
C	25	0	0	1	25	
C	29	0	0	1	29	

ORDINAL ENCODING

Ordinal

Rate		Rate
Great	OrdinalEncoder → 	4
OK		2
Great		4
Terrible		0
Bad		1
Great		4
Good		3

NIVELES DE MEDIDA

DATOS NOMINALES

$$A \neq B$$

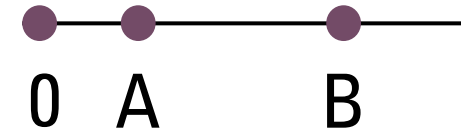
DATOS DE INTERVALO



DATOS ORDINALES

$$A < B$$

DATOS DE RAZON



NIVELES DE MEDIDA

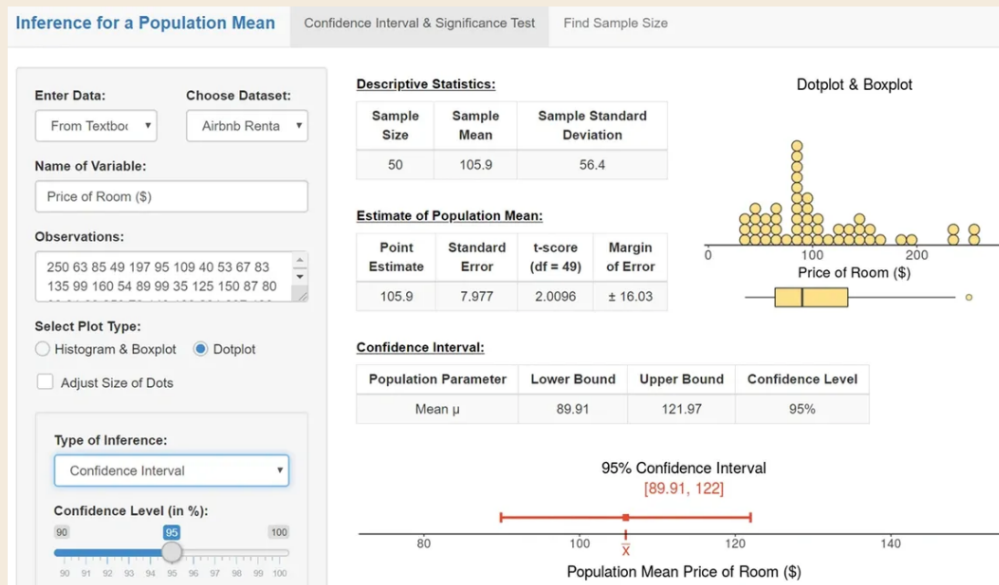
<https://view.genially.com/6434ba894201cc001a5e9944/interactive-content-cuadro-comparativo-escalas-de-medicion-de-datos>

CUADRO COMPARATIVO ESCALAS DE MEDICIÓN DE LOS DATOS					
	DEFINICIÓN	VARIABLES	CARACTERÍSTICAS	GRAFICOS	EJEMPLO
NOMINAL	MEDICIÓN QUE SIRVE COMO ETIQUETA PARA IDENTIFICAR O CLASIFICAR UN OBJETO	NO NUMÉRICAS (NO CUANTITATIVAS) ARTÍCULOS, OBJETOS O ELEMENTOS NO ORIENTADOS A UN NÚMERO (SOLO SE PUEDEN CLASIFICAR EN CLASES)	- SIRVE PARA IDENTIFICAR, DAR UN VALOR, CORRELACIONAR, DESCRIBIR, ETC. - SE DIVIDE EN AL MENOS 3 CATEGORÍAS PARA PODER MEDIR - NATURALEZA CUALITATIVA - LOS NÚMEROS NO DEFINEN LAS CARACTERÍSTICAS	 GRÁFICA DE BARRAS DIAGRAMA DE SECTORES PICTOGRAMAS	- GÉNERO - ESTADO CIVIL - PROFESIÓN - COLOR - SI TIENES ALGO O NO
ORDINAL	CLASIFICA Y ORDENA LOS DATOS ESTADÍSTICOS QUE TIENEN LA MISMA NATURALIDAD PERO HAY DIFERENCIA ENTRE ELLOS	MENORES O MAYORES (CLASIFICACIÓN INTRÍNSECA) CUALITATIVA	- PUEDEN SER ORDENAS POR JERARQUÍA (ASCENDENTE Y DESENDENTE) - SIRVEN PARA LA OBTENCIÓN DE DATOS - IDENTIFICA EL RANGO DE LAS VARIABLES	 GRÁFICA DE BARRAS ESCALA DE LIKERT	- CLASIFICACIÓN CREDITICIA - CALIFICACIONES DE LOS EXÁMENES - POSICIÓN EN UNA CARRERA - LA FRECUENCIA EN QUE PASA ALGO
INTERVALO	ESCALA DE MEDICIÓN QUE MIDE LA DIFERENCIA ENTRE DOS VARIABLES (RESTA O SUMA ENTRE LAS DOS VARIABLES)	CUANTITATIVA CON VALORES REALES DONDE EL CERO ES ARBITRARIO.	- USA NUMEROS ENTEROS O FRACCIONARIOS - PERMITE CALCULAR LA MEDIA DE LAS VARIABLES - PERMITE ASIGNAR VALOR NUMÉRICO A CUALQUIER EVALUACIÓN - CAPAZ DE CUANTIFICAR Y DIFERENCIAR ENTRE OPCIONES - PERMITE ANALIZAR UNA GRAN CANTIDAD DE DATOS	 ESCALA DE LIKERT MATRIZ DE ESCALA BIPOLAR NET PROMOTER SCORE (NPS)	- TEMPERATURA EN GRADOS - NIVELES DE SATISFACCIÓN O RECOMENDACIÓN
RAZÓN	ESCALA DE MEDIDA O MODO DE MEDICIÓN QUE PERMITE HACER LA COMPARACIÓN DEL TIPO CUANTAS VECES ES MAYOR UN ELEMENTO QUE OTRO	VARIABLES CUANTITATIVAS CON PUNTO DE CERO ABSOLUTO (NO HAY VALORES NUMÉRICOS NEGATIVOS)	- LOS VALORES PUEDEN SER SUMADOS, RESTADOS, MULTIPLICADOS O DIVIDIDOS - OFRECE TODAS LAS FUNCIONES DE LAS OTRAS 3 ESCALAS	 TABLA CRUZADA FODA ANÁLISIS TUF	- ALTURA - ESTATURA - PESO - TIEMPO - EDAD - INGRESOS - COSTOS - CANTIDAD DE ALGO - SALARIO

NIVELES DE MEDIDA

[Home](#)[Web Apps](#)[Mobile Apps](#)

THE ART & SCIENCE OF LEARNING FROM DATA

[Datasets](#)[R Code](#)[YouTube](#)[Errata](#)

Art of Stat

Web Apps

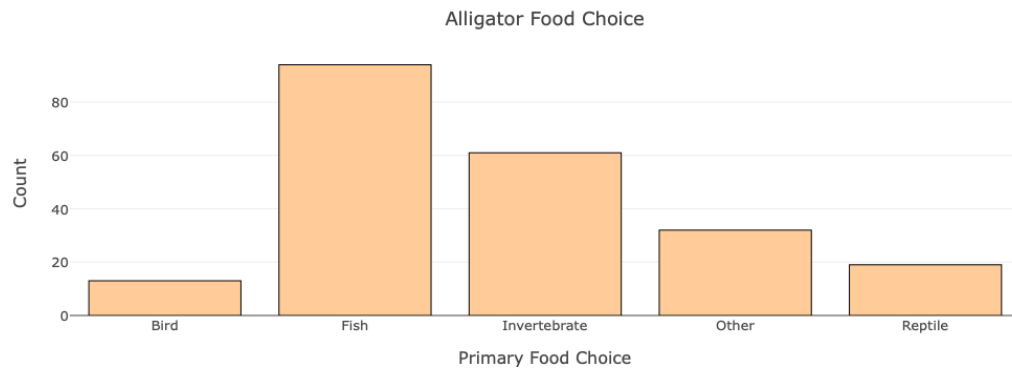
Explore statistical concepts in an interactive way. Use the apps to construct graphs, obtain summary statistics, find probabilities, get confidence intervals or fit linear regression models. Take screenshots or download graphs of your data.

[Overview of Web Apps](#) >

<https://artofstat.com/web-apps>

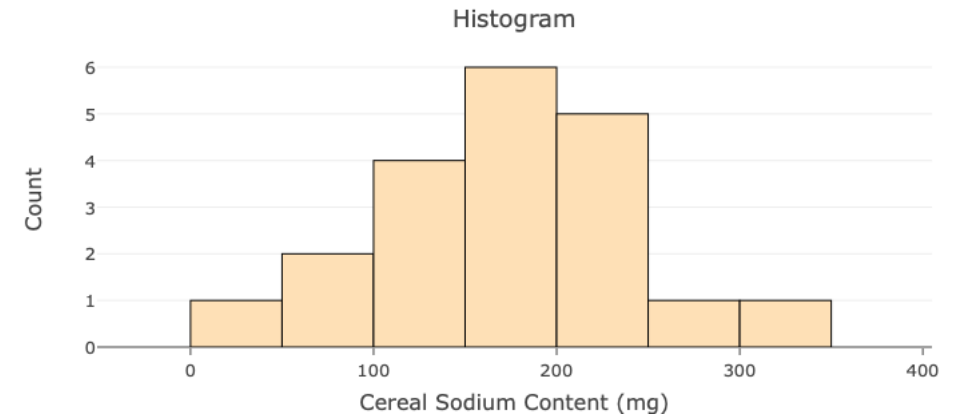
NIVELES DE MEDIDA

DATOS CATEGÓRICOS



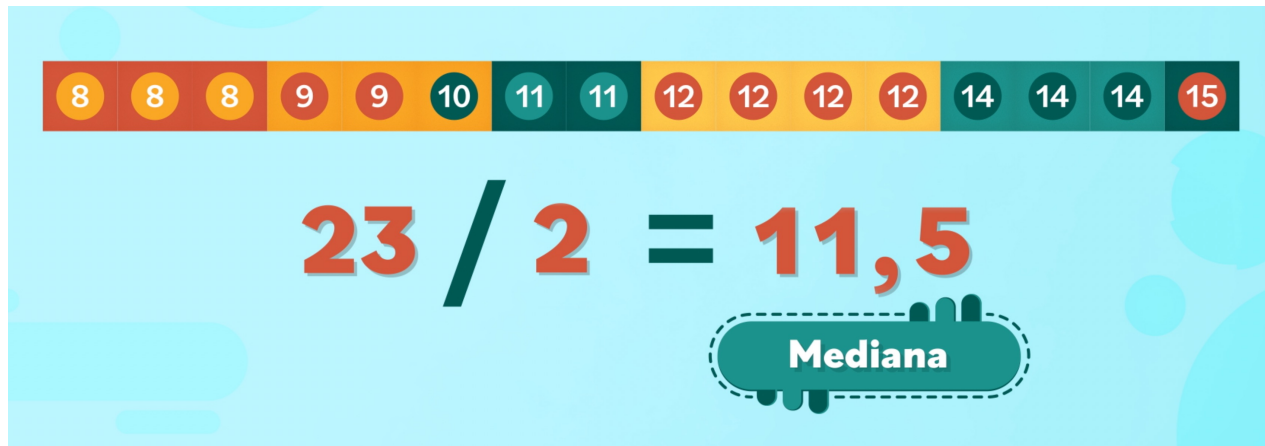
https://istats.shinyapps.io/EDA_categorical/

DATOS CUANTITATIVOS



https://istats.shinyapps.io/EDA_quantitative/

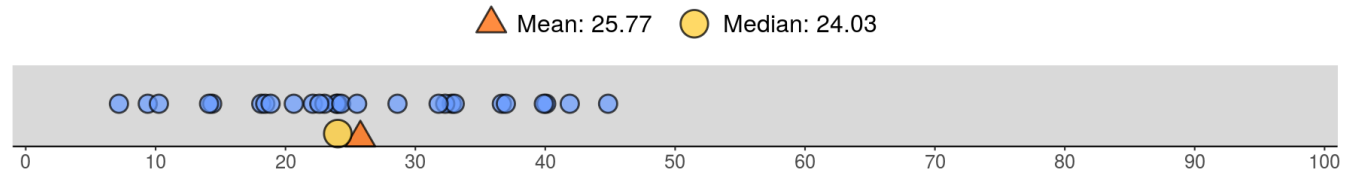
PROMEDIO, MEDIANA Y MODA



PROMEDIO, MEDIANA Y MODA

MEAN VS MEDIAN

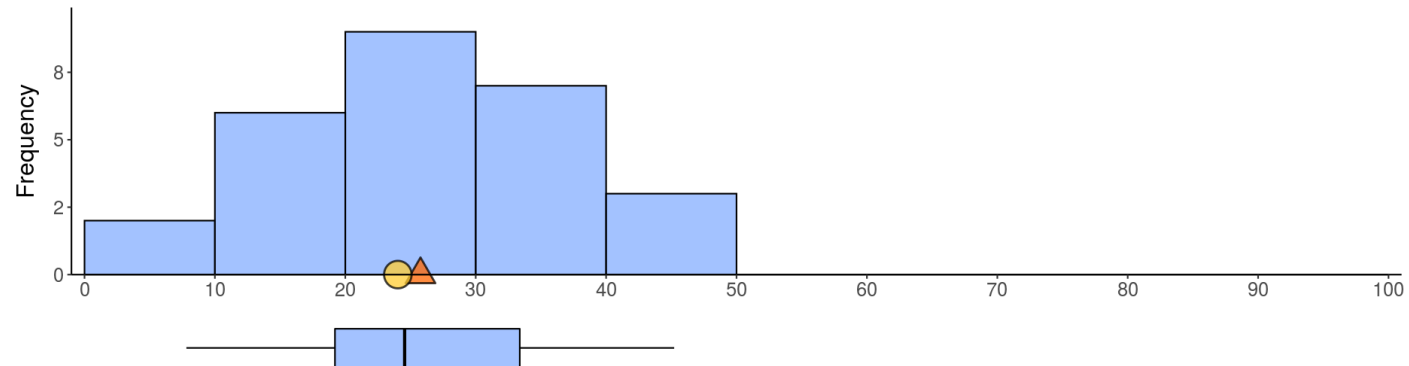
You can add or delete points by clicking in the gray area.



Descriptive Statistics:

Sample Size	Mean	Standard Deviation	Minimum	1st Quartile	Median	3rd Quartile	Maximum	IQR
27	25.77	10.39	7.18	18.65	24.03	32.92	44.84	14.27

Histogram with Boxplot



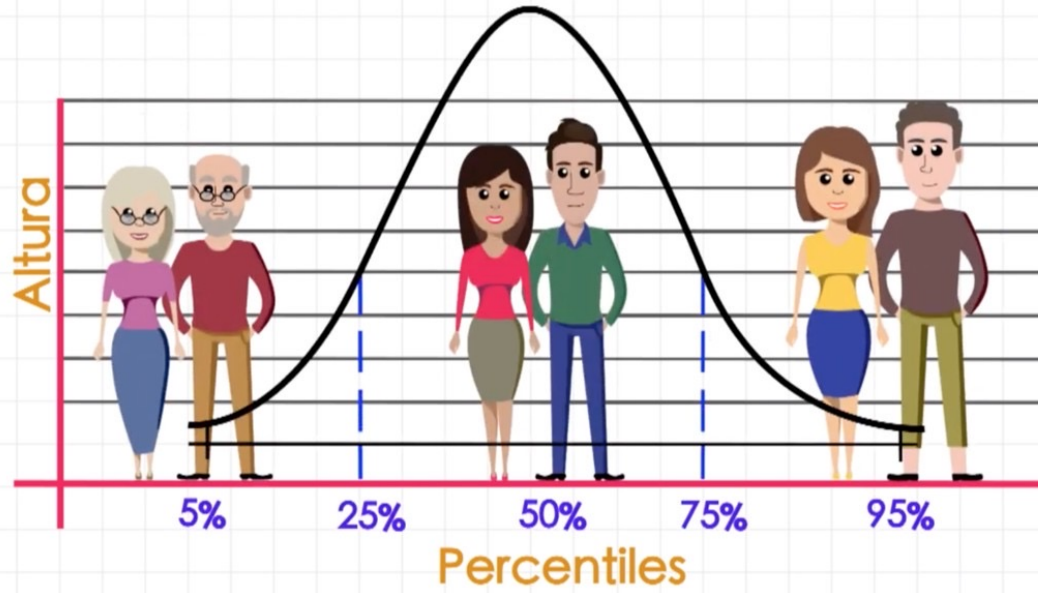
<https://ista-ts.shinyapps.io/MeanvsMedian/>

PROMEDIO, MEDIANA Y MODA

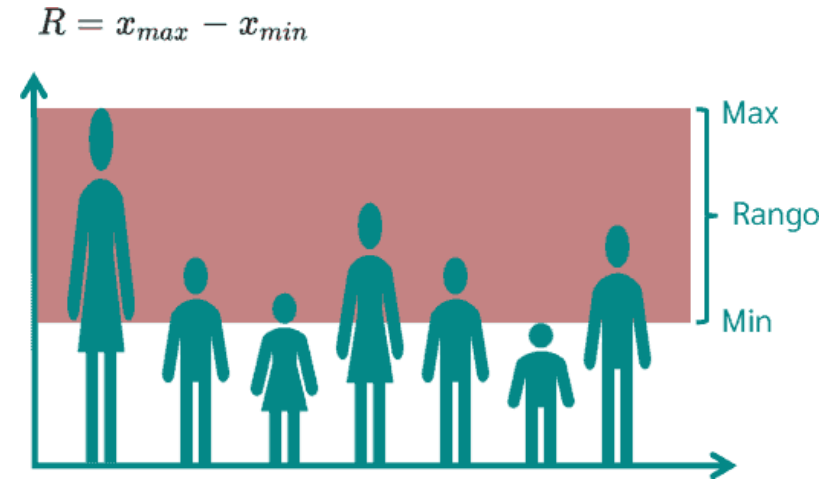
Provides:	Nominal	Ordinal	Interval	Ratio
The "order" of values is known		✓	✓	✓
"Counts," aka "Frequency of Distribution"	✓	✓	✓	✓
Mode	✓	✓	✓	✓
Median		✓	✓	✓
Mean			✓	✓
Can quantify the difference between each value			✓	✓
Can add or subtract values			✓	✓
Can multiple and divide values				✓
Has "true zero"				✓

<https://datascience.eu/es/matematica-y-estadistica/tipos-de-datos-y-escalas-de-medicion-nominal-ordinal-intervalo-y-razon/>

RANGO, VARIANZA Y DESVIACIÓN ESTÁNDAR



RANGO



RANGO, VARIANZA Y DESVIACIÓN ESTÁNDAR

CUANTILES, CUARTILES, ETC



RANGO, VARIANZA Y DESVIACIÓN ESTÁNDAR

RANGO

Rango



Distancia entre el valor más bajo y el más alto de una distribución.

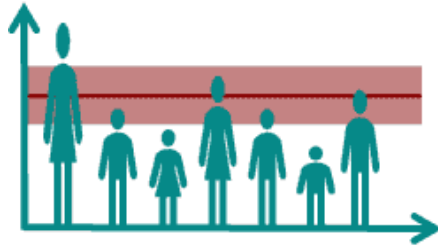
Rango intercuartílico



Espectro en el que se sitúa el 50% medio de los valores. Diferencia entre el primer y el tercer cuartil.

RANGO, VARIANZA Y DESVIACIÓN ESTÁNDAR

Desviación estándar



Distancia media de todos los valores medidos al valor medio.

Rango

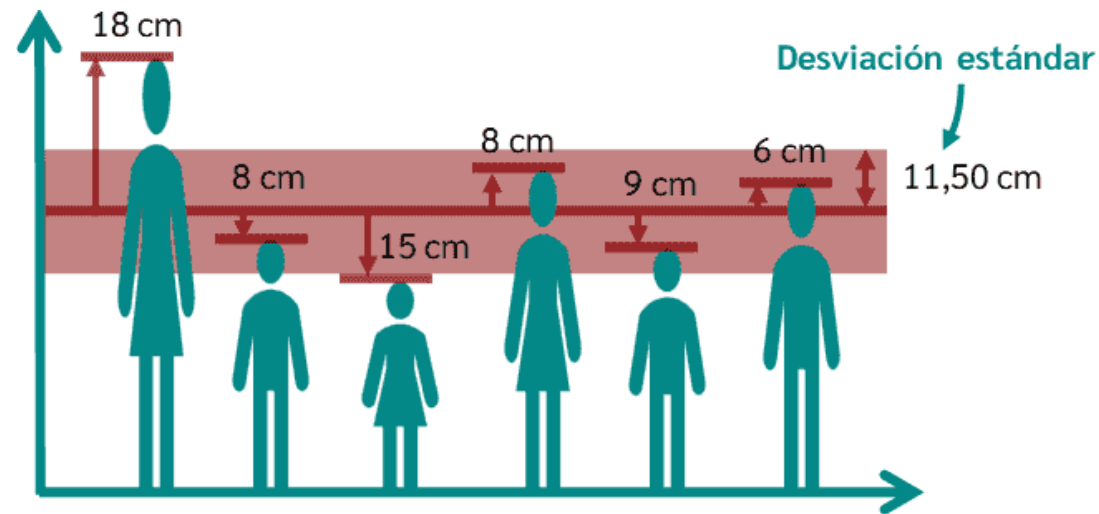


Distancia entre el valor más bajo y el más alto de una distribución.

Rango intercuartílico



Espectro en el que se sitúa el 50% medio de los valores. Diferencia entre el primer y el tercer cuartil.



VARIANZA

<https://seeing-theory.brown.edu/basic-probability/index.html#section3>

☰ Seeing Theory

Chapter 1

Basic Probability

This chapter is an introduction to the basic concepts of probability theory.



Chance Events



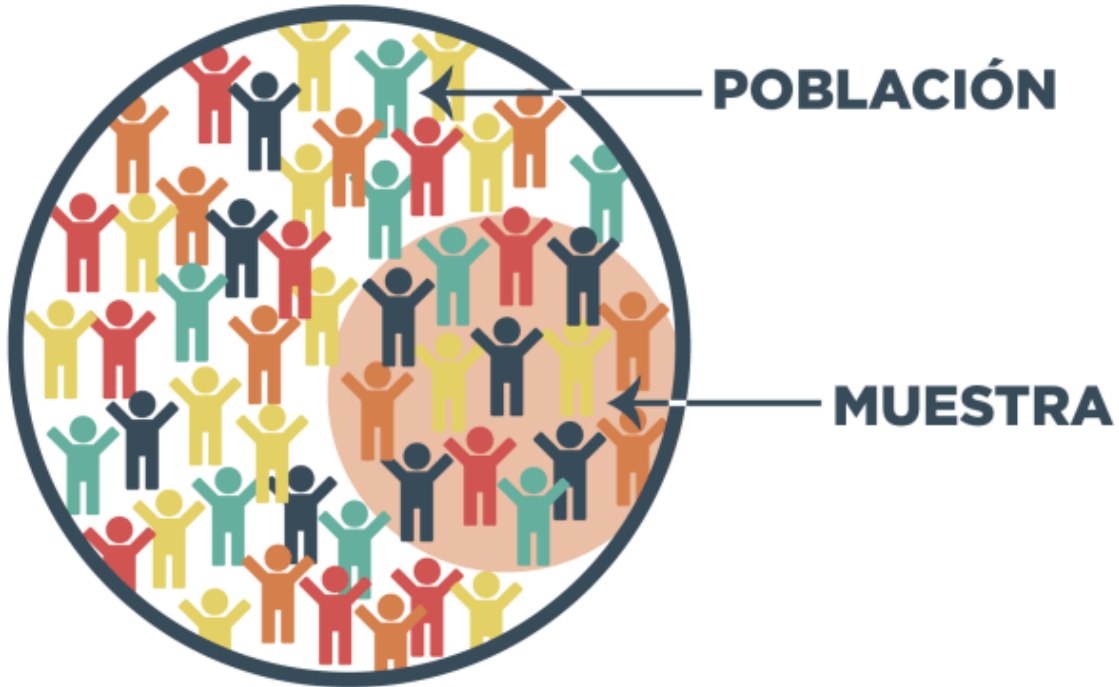
Expectation



Variance



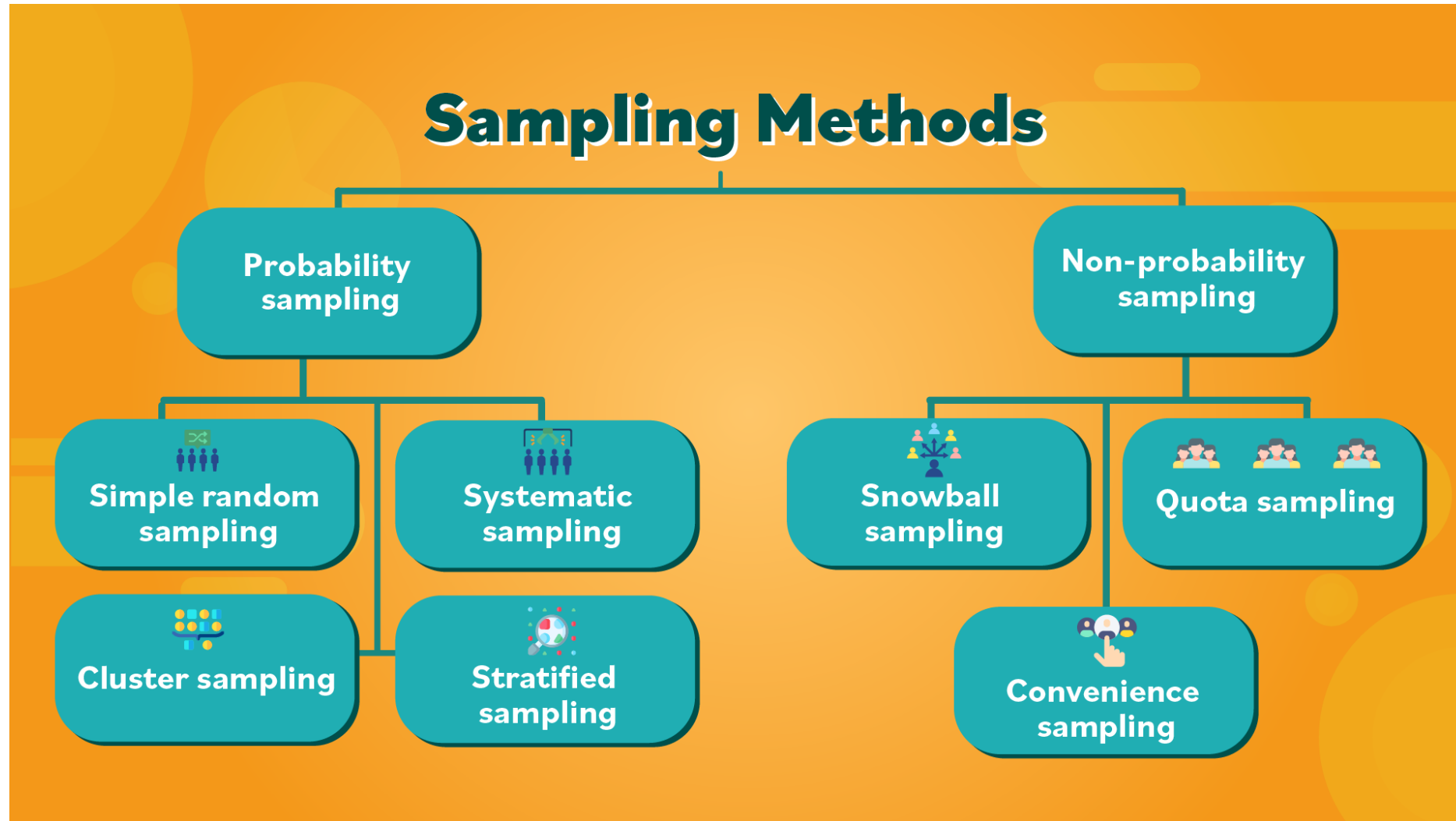
POBLACIÓN Y MUESTRA



**A GOOD SAMPLE IS
ONE THAT IS REPRESENTATIVE OF THE ENTIRE POPULATION**

<https://www.youtube.com/watch?v=pTuj57uXW1k>

POBLACIÓN Y MUESTRA

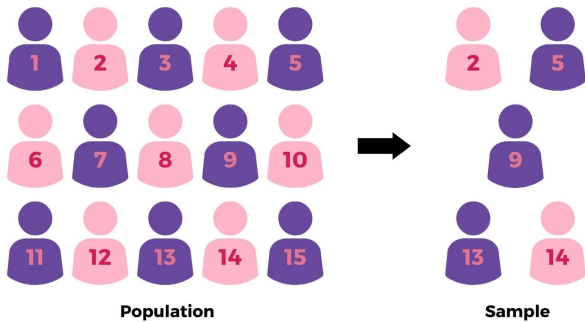


POBLACIÓN Y MUESTRA

MUESTREO ALEATORIO SIMPLE



Simple Random Sampling



<https://tgmresearch.com/simple-random-sampling.html>

MUESTREO SISTEMÁTICO



<https://edu.gcfglobal.org/en/statistics-basic-concepts/what-is-probability/1/>

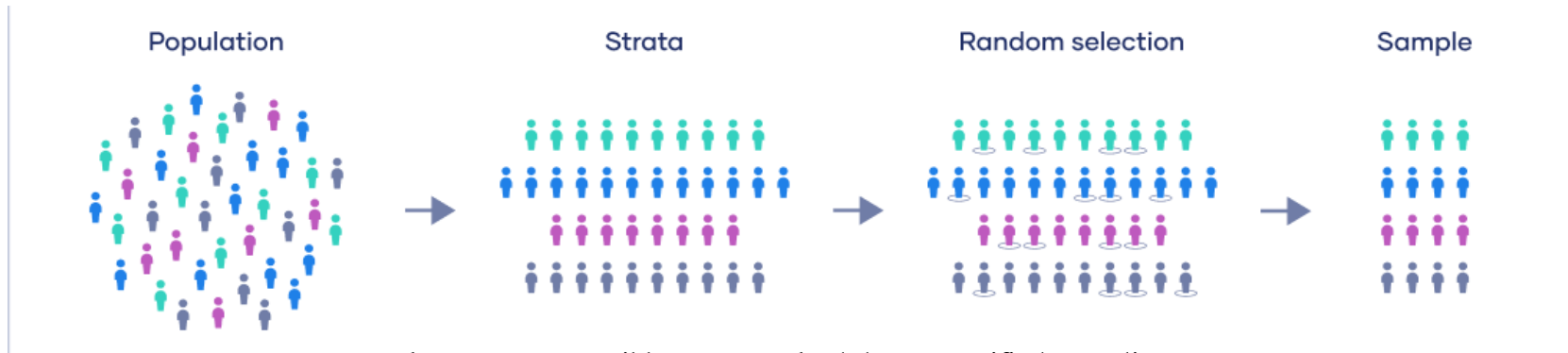
POBLACIÓN Y MUESTRA

MUESTREO POR CONGLOMERADOS



<https://www.scribbr.com/methodology/cluster-sampling/>

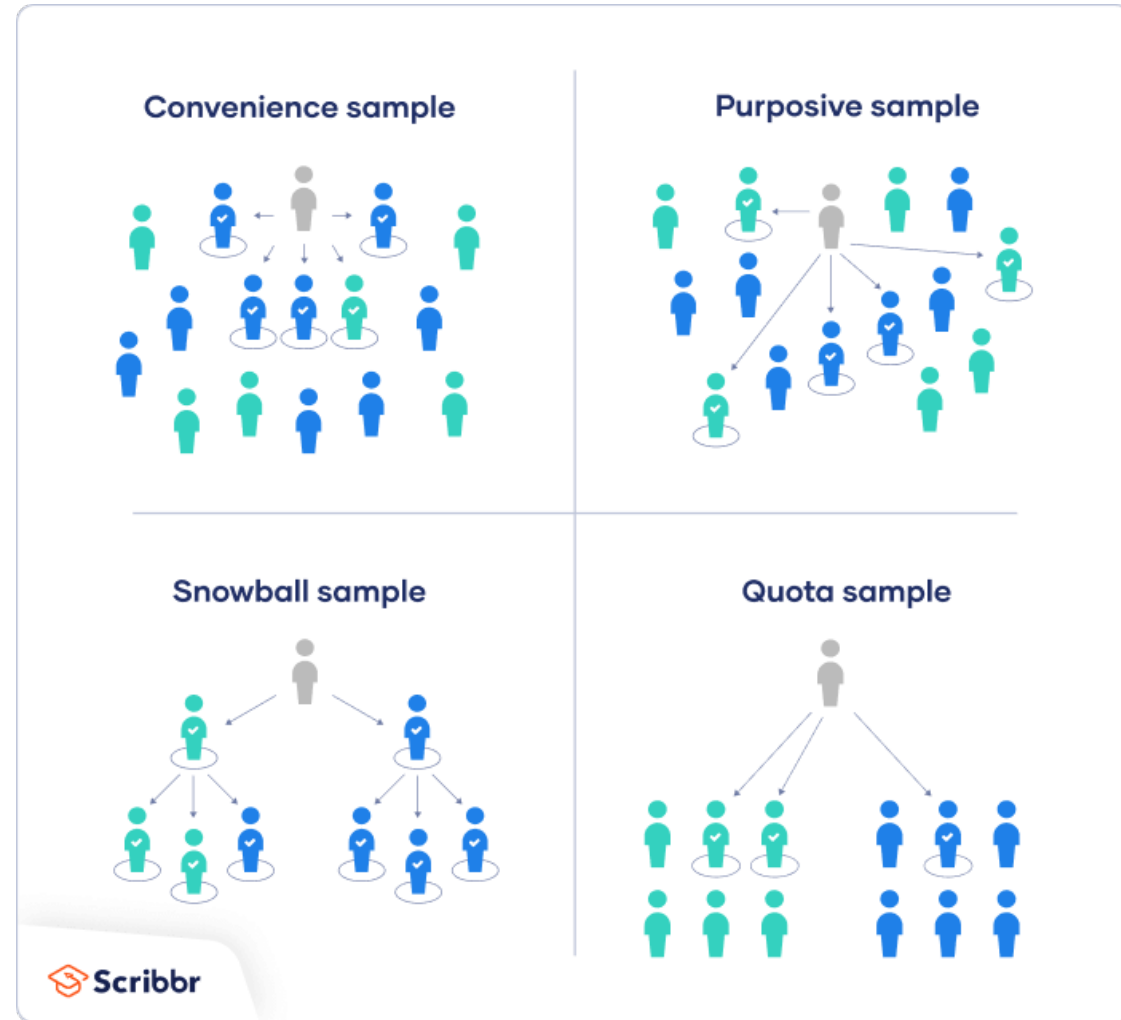
MUESTREO ESTRATIFICADO



<https://www.scribbr.com/methodology/stratified-sampling/>

POBLACIÓN Y MUESTRA

MÉTODOS NO PROBABILÍSTICOS



<https://www.scribbr.com/methodology/sampling-methods/>

PROBABILIDAD



Chapter 1: Basic Probability

Chance Events

Randomness is all around us. Probability theory is the mathematical framework that allows us to analyze chance events in a logically sound manner. The probability of an event is a number indicating how likely that event will occur. This number is always between 0 and 1, where 0 indicates impossibility and 1 indicates certainty.

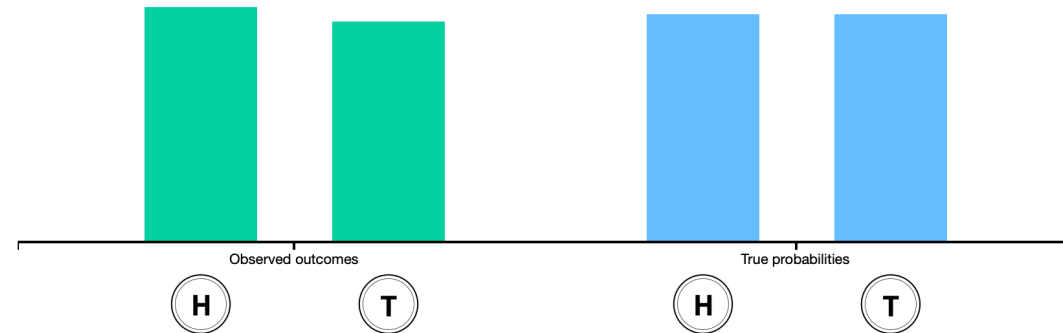
A classic example of a probabilistic experiment is a fair coin toss, in which the two possible outcomes are heads or tails. In this case, the probability of flipping a head or a tail is $1/2$. In an actual series of coin tosses, we may get more or less than exactly 50% heads. But as the number of flips increases, the long-run frequency of heads is bound to get closer and closer to 50%.



Flip the Coin

Flip 100 times

For an unfair or weighted coin, the two outcomes are not equally likely. You can change the weight or distribution of the coin by dragging the true probability bars (on the right in blue) up or down. If we assign numbers to the outcomes — say, 1 for heads, 0 for tails — then we have created the mathematical object known as a [random variable](#).



<https://seeing-theory.brown.edu/basic-probability/index.html#section3>

PROBABILIDAD

TEORÍA DE CONJUNTOS

 Chapter 2: Compound Probability

Set Theory

A set, broadly defined, is a collection of objects. In the context of probability theory, we use set notation to specify compound events. For example, we can represent the event "roll an even number" by the set {2, 4, 6}. For this reason it is important to be familiar with the algebra of sets.

Use the set constructor below to build a set, then press "Submit" to see your set visualized in the Venn diagram. You can also move and resize the circles by dragging and dropping.

A B C $\emptyset$ U

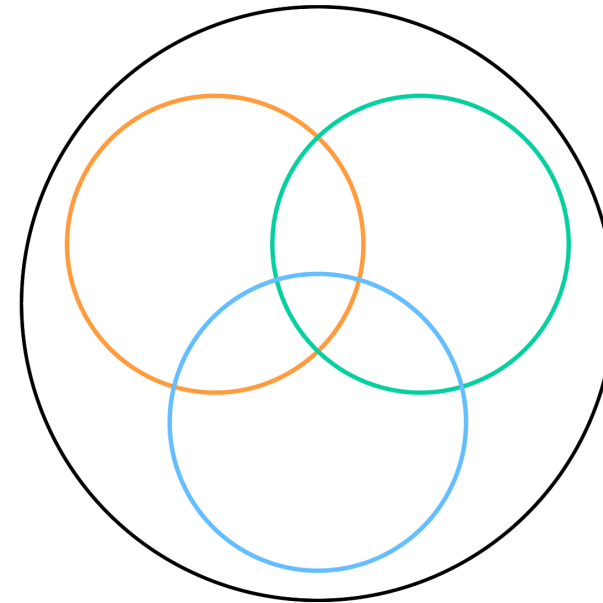
(n u ')

Submit

Delete

Reset

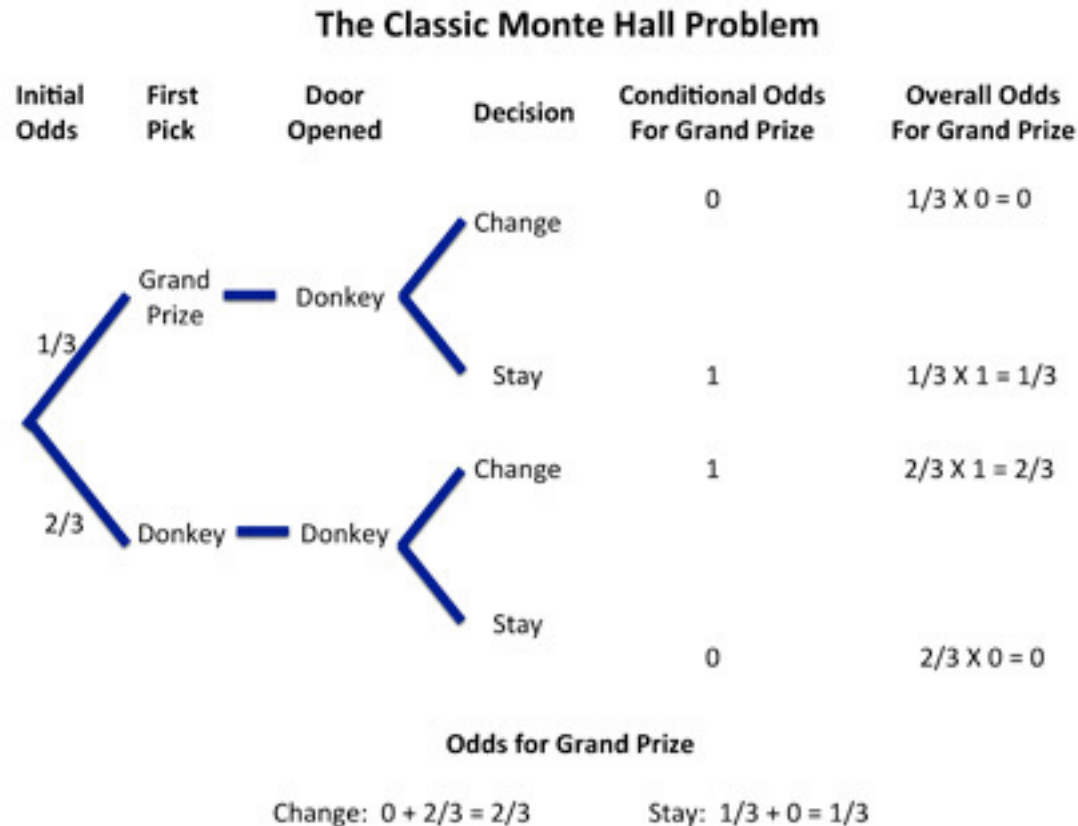
A
B
C



<https://seeing-theory.brown.edu/compound-probability/index.html#section1>

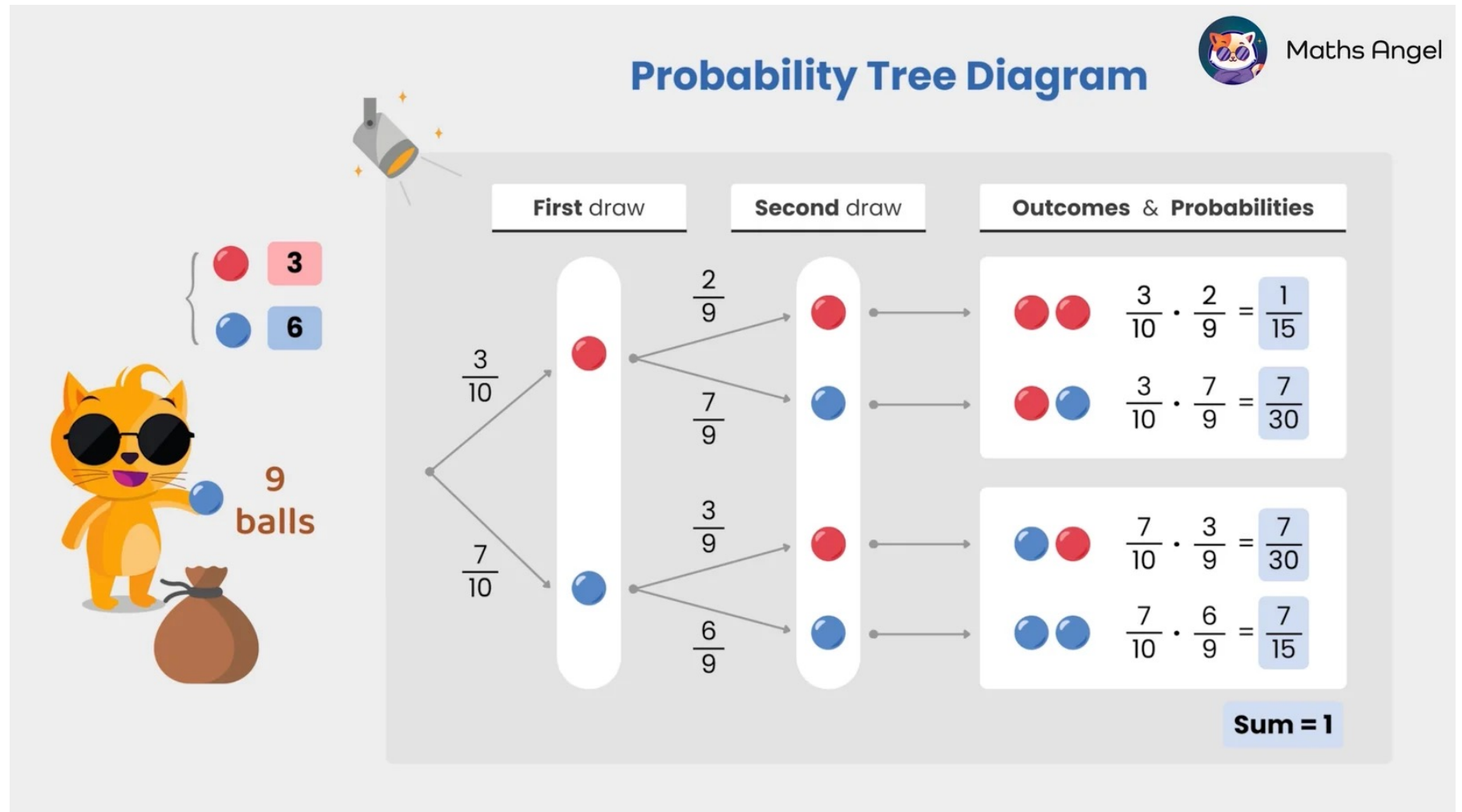
PROBABILIDAD

EL PROBLEMA DE MONTY HALL



PROBABILIDAD

EVENTOS COMPUESTOS



PROBABILIDAD



Types of events in probability

Independent events is an events that are **not affected** by other events.



Two dice toss at the same time will have the same chance of outcome

Dependent events is an events that **can be affected** by previous events



Tossing the dice first time and then toss that again will have a different chance of outcome

Mutually exclusive events is an events that **cannot occur** at the same time



Can you toss one dice with two different outcome? No, because they cannot happen at the same time

PROBABILIDAD

PROBABILIDAD CONDICIONAL

 Chapter 2: Compound Probability

Conditional Probability

Conditional probabilities allow us to account for information we have about our system of interest. For example, we might expect the probability that it will rain tomorrow (in general) to be smaller than the probability it will rain tomorrow *given that it is cloudy today*. This latter probability is a conditional probability, since it accounts for relevant information that we possess.

Mathematically, computing a conditional probability amounts to shrinking our sample space to a particular event. So in our rain example, instead of looking at how often it rains on any day in general, we "pretend" that our sample space consists of only those days for which the previous day was cloudy. We then determine how many of those days were rainy.

Click on the tabs below to visualize the shrinking of the sample space.

A

B

C

Universe

<https://seeing-theory.brown.edu/compound-probability/index.html#section3>

Conditional probability

Explained Visually

 Post  Me gusta 26  Compa

By [Victor Powell](#)

A conditional probability is the probability of an event, given some other event has already occurred. In the below example, there are two possible events that can occur. A ball falling could either hit the **red** shelf (we'll call this event **A**) or hit the **blue** shelf (we'll call this event **B**) or both.

If we know the statistics of these events across the entire population and then were to be given a single ball and told "this ball hit the **red** shelf (event **A**), what's the probability it also hit the **blue** shelf (event **B**)?" we could answer this question by providing the conditional probability of **B** given that **A** occurred or $P(B|A)$.

$$P(B|A) = \frac{P(A \cap B)P(B)}{P(A)}.$$

<https://setosa.io/ev/conditional-probability/>

Markov Chains

Explained Visually

🔗 Post 👍 Me gusta 231 📄 Comp

By [Victor Powell](#)

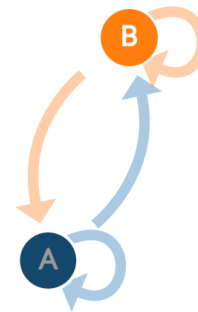
with text by [Lewis Lehe](#)

CADENAS DE MARKOV

Markov chains, named after [Andrey Markov](#), are mathematical systems that hop from one "state" (a situation or set of values) to another. For example, if you made a Markov chain model of a baby's behavior, you might include "playing," "eating," "sleeping," and "crying" as states, which together with other behaviors could form a 'state space': a list of all possible states. In addition, on top of the state space, a Markov chain tells you the probability of hopping, or "transitioning," from one state to any other state---e.g., the chance that a baby currently playing will fall asleep in the next five minutes without crying first.

A simple, two-state Markov chain is shown below.

 speed



<https://setosa.io/ev/markov-chains/>

PROBABILIDAD



Chapter 5: Bayesian Inference

Bayes' Theorem

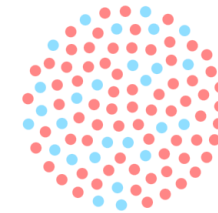
Suppose that on your most recent visit to the doctor's office, you decide to get tested for a rare disease. If you are unlucky enough to receive a positive result, the logical next question is, "Given the test result, what is the probability that I actually have this disease?" (Medical tests are, after all, not perfectly accurate.) Bayes' Theorem tells us exactly how to compute this probability:

$$P(\text{Disease}|+) = \frac{P(+|\text{Disease})P(\text{Disease})}{P(+)}$$

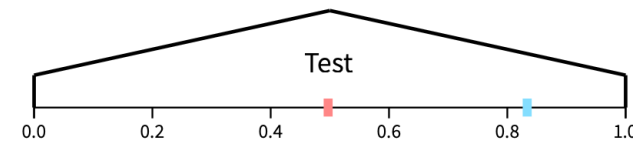
As the equation indicates, the *posterior* probability of having the disease given that the test was positive depends on the *prior* probability of the disease $P(\text{Disease})$. Think of this as the incidence of the disease in the general population. Set this probability by dragging the bars below.



TEOREMA DE BAYES



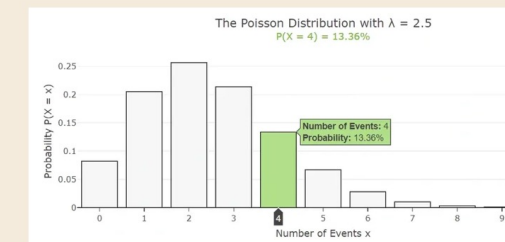
Population



Negative

Positive

DISTRIBUCIONES DE PROBABILIDAD



PROBABILIDAD

OTRA APLICACIÓN DE LA PROBABILIDAD BAYESIANA

 Chapter 5: Bayesian Inference

Prior to Posterior

At the core of Bayesian statistics is the idea that prior beliefs should be updated as new data is acquired. Consider a possibly biased coin that comes up heads with probability p . This purple slider determines the value of p (which would be unknown in practice).

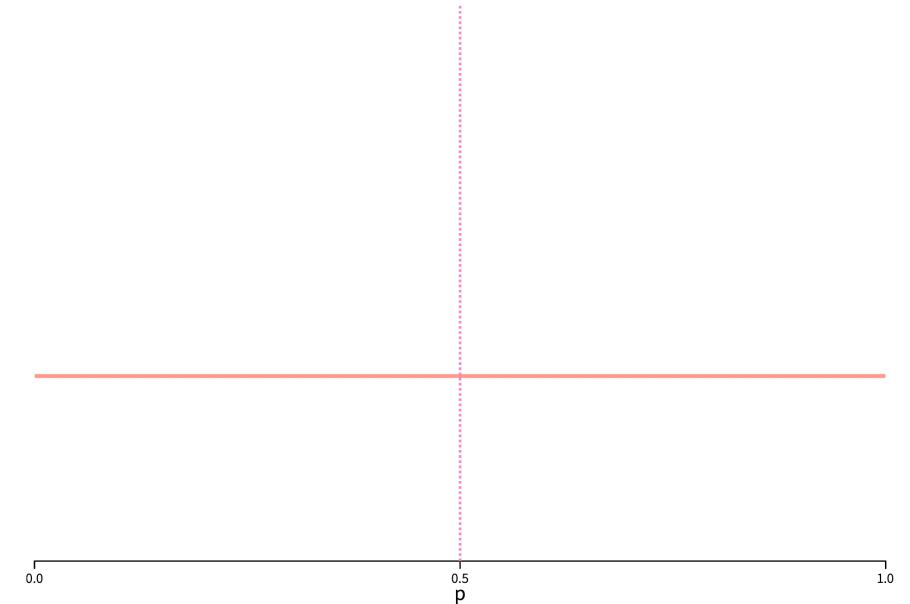
$p = 0.5$ 

The pink sliders control the shape of the initial $\text{Beta}(\alpha, \beta)$ prior distribution, the density function of which is also plotted in pink.

$\alpha = 1$ 

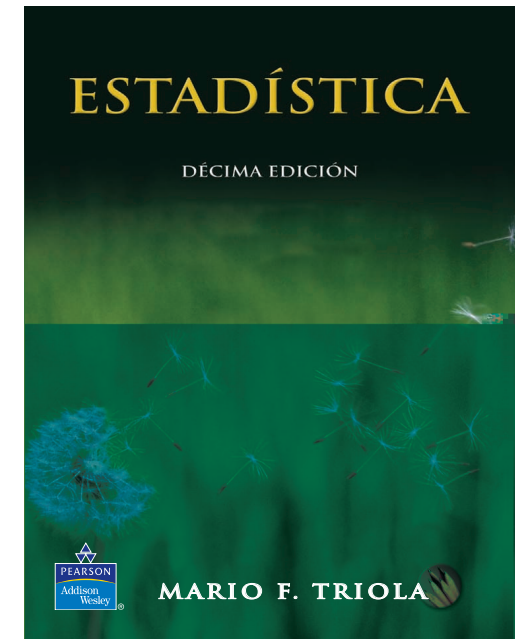
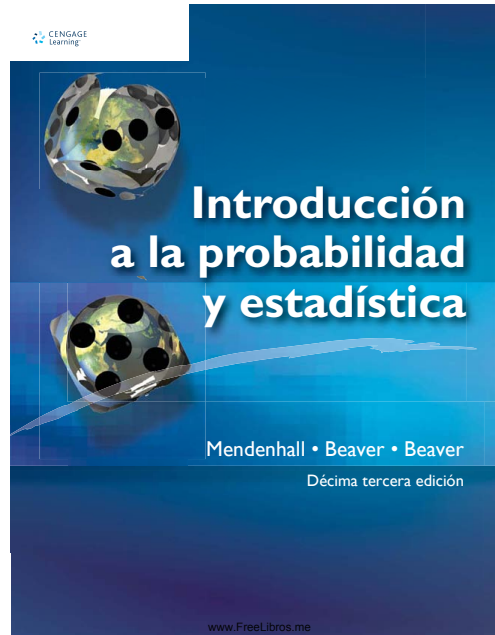
$\beta = 1$ 

As we acquire data in the form of coin tosses, we update the posterior distribution on p , which represents our best guess about the likely values for the bias of the coin. This updated distribution then serves as the prior for future coin tosses.



<https://seeing-theory.brown.edu/bayesian-inference/index.html#section2>

RECOMENDACIONES

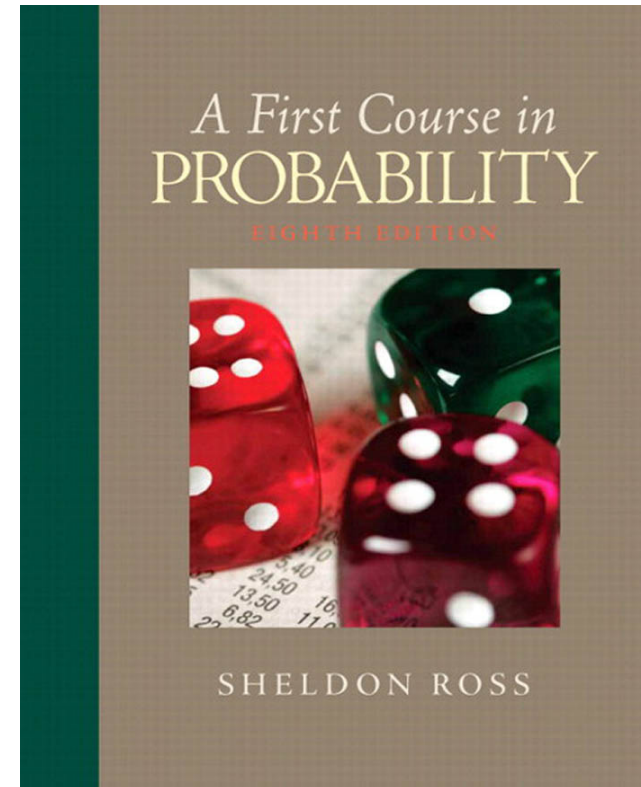


RECOMENDACIONES

Introducción a la probabilidad

Luis Rincón
Departamento de Matemáticas
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Circuito Exterior de CU
04510 México DF

Agosto 2014



RECOMENDACIONES

